

$\begin{cases} \text{PCA} \\ \text{NB} \end{cases}$

ml  $\leadsto$  edu.

# Deep Learning



what is ?

بہتر عاقل / DL / Deep Learning کیا ہے؟

Voice  
Text  
Image

## ARTIFICIAL INTELLIGENCE VS MACHINE LEARNING VS DEEP LEARNING

### 1 Artificial Intelligence

Development of smart systems and machines that can carry out tasks that typically require human intelligence

Traditional → if/else

### 2 Machine Learning

Creates algorithms that can learn from data and make decisions based on patterns observed  
Require human intervention when decision is incorrect

Logistic Regression

Linear

Support Vector

Random Forest

Decision Tree

Naive Bayes

Probabilistic

Tabular

### 3 Deep Learning

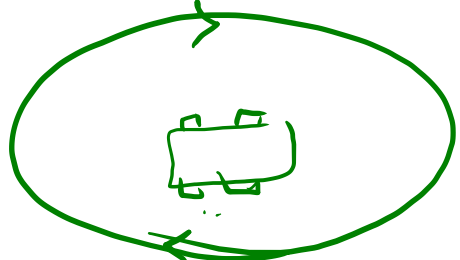
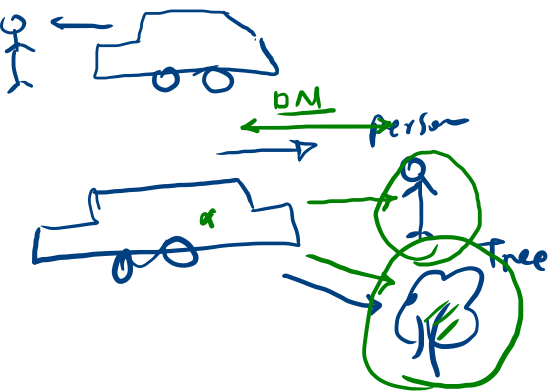
Uses an artificial neural network to reach accurate conclusions without human intervention

Neural Network



Deep learning

تجزیه و تحلیل



object Det.

تشخیص

Distance measurement

تجزیه و تحلیل

road segmentation

car out.

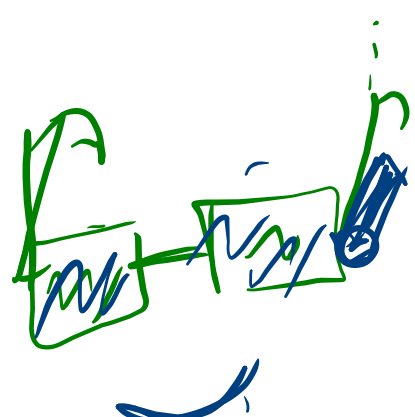
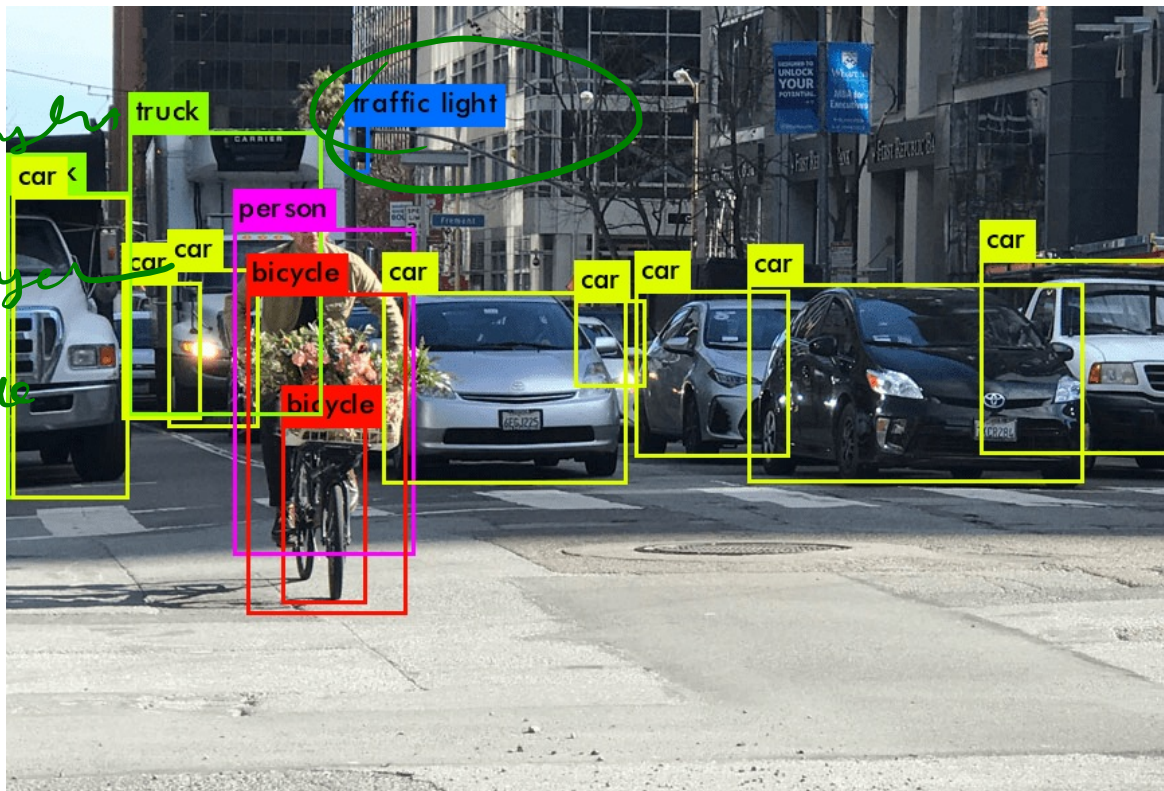
object out

traffic light

car

bicycle

Truck



Describe the scene!

{ face Det.  
face recog.





trained

Choose base model

Label data

Training

AI Assistant

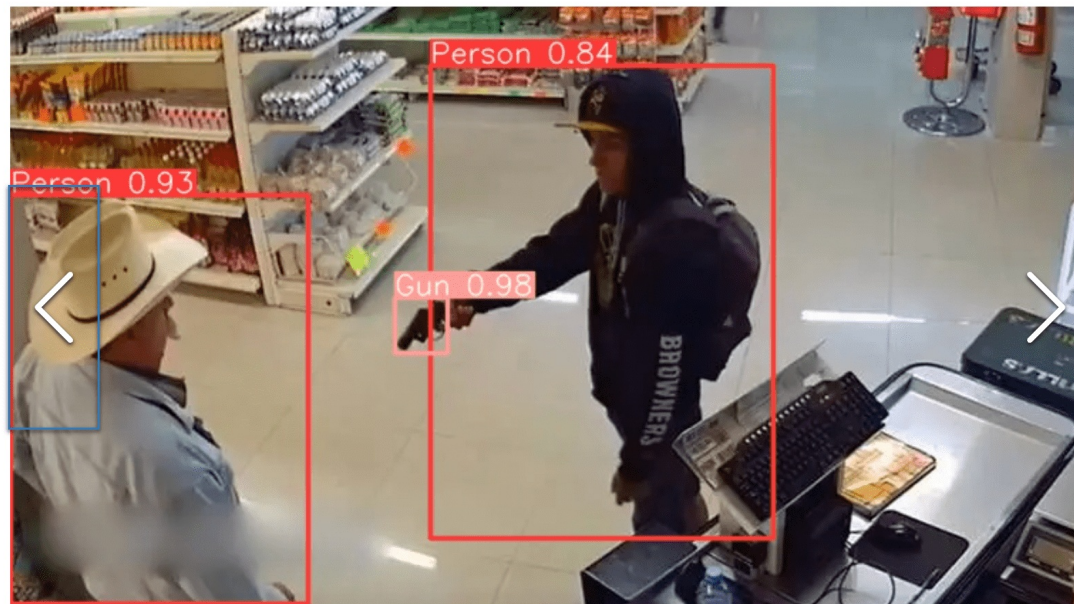
Welcome to VisionPlatform!  
How can I assist you today?

Type your question



## Training results

## Test Model - View Predictions



Back

Done

Inference resolution

640x640

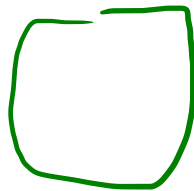
Classes

PERSON

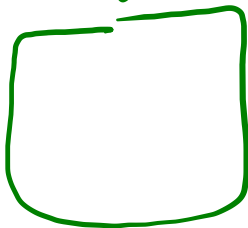
GUN

Gpt / Gemini / Claude / ...

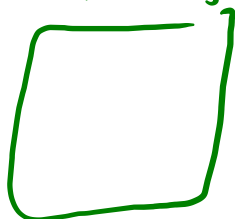
winter



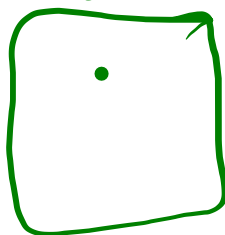
summer



young



old

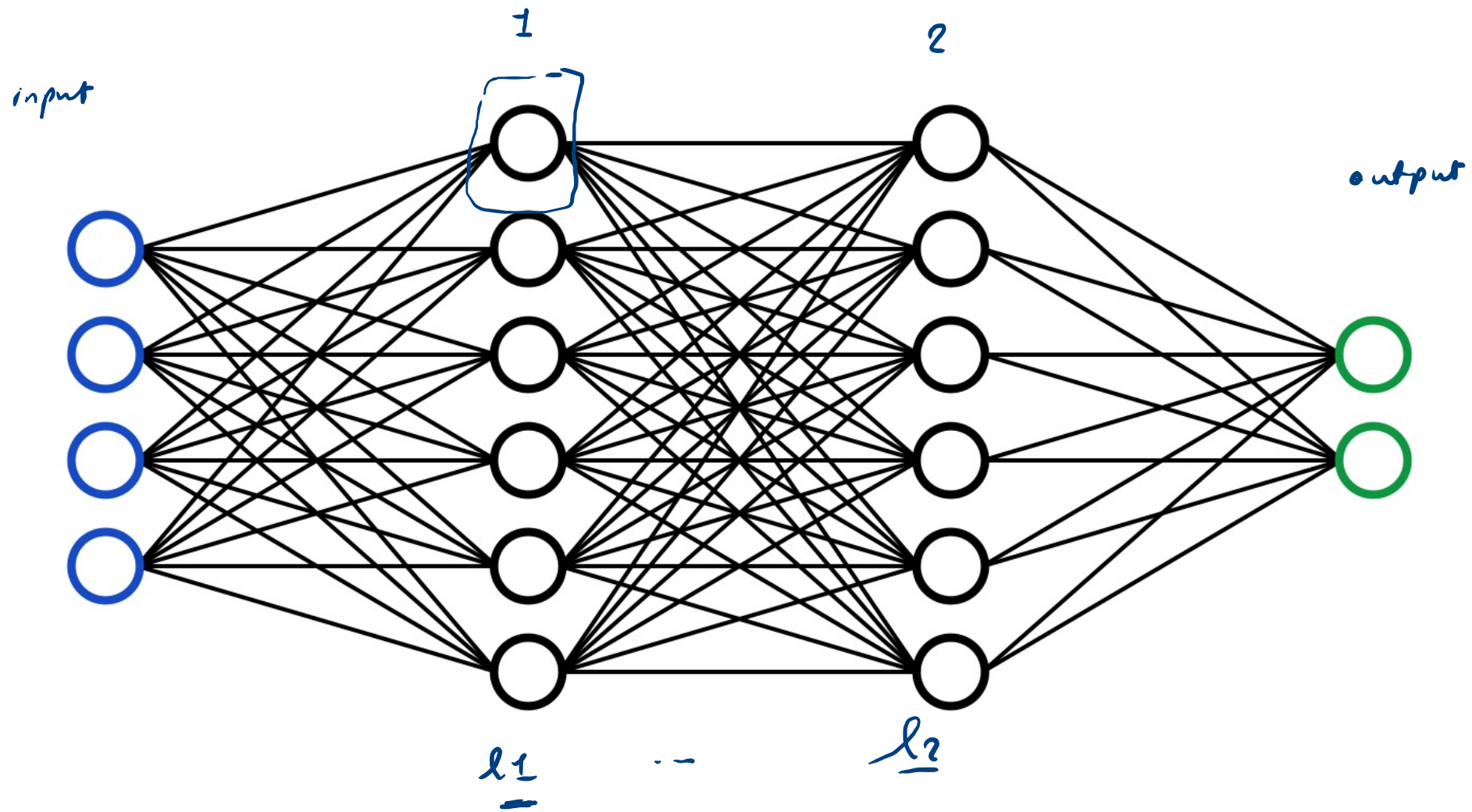


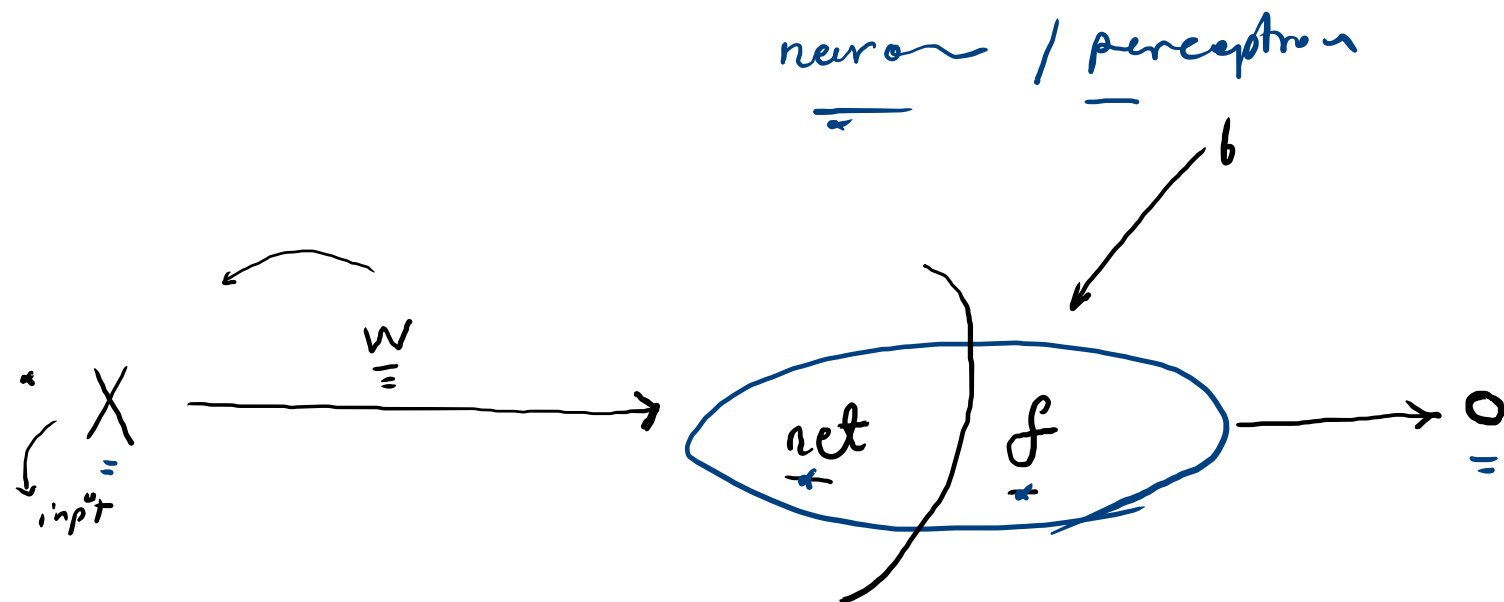


(neural nets) ? ~~improvement~~

---

a





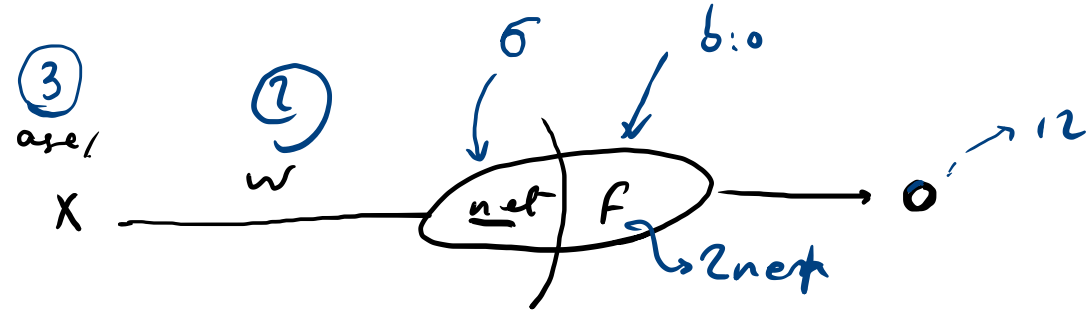
$$\text{net: } W X + b$$

$$O = f(\text{net})$$

$\downarrow$   
*using*  
 Activation Function

House price Estimation  $\rightarrow \frac{NN}{a}$

| age | material | price |
|-----|----------|-------|
| 3   | A        | —     |
| 9   | B        | —     |
| 8   | S        | —     |



$$\underline{w = 2}$$

$$x = 3$$

$$b = 0$$

$$\rightarrow net = wx + b = 2 \times 3 + 0 = 6$$

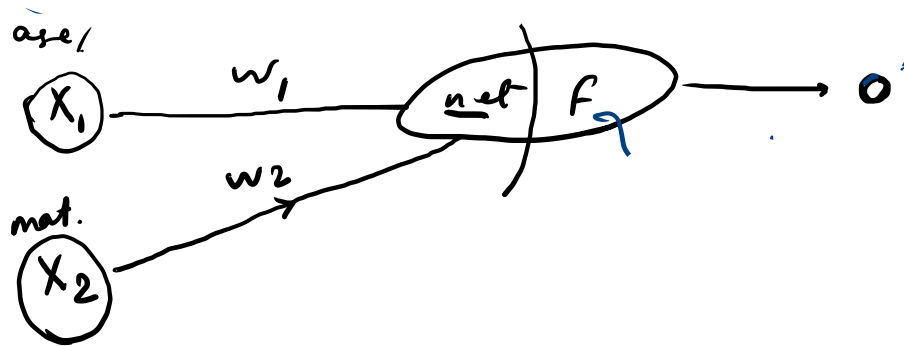
$$\alpha \quad f = \underline{2net} \rightarrow o = f(net) = 2net$$

$$o = 2 \times 6 = 12$$

House price Estimation  $\rightarrow \frac{NN}{n}$

$\rightarrow$  ↓

| <u>age</u> | material | price |
|------------|----------|-------|
| 3          | A        | —     |
| 9          | B        | —     |
| 8          | S        | —     |

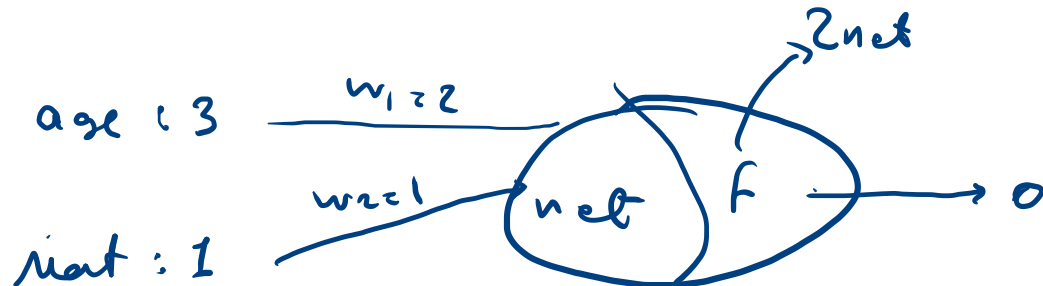


$$O = f(net) = 2net = 2 \times 7 = \underline{14}$$

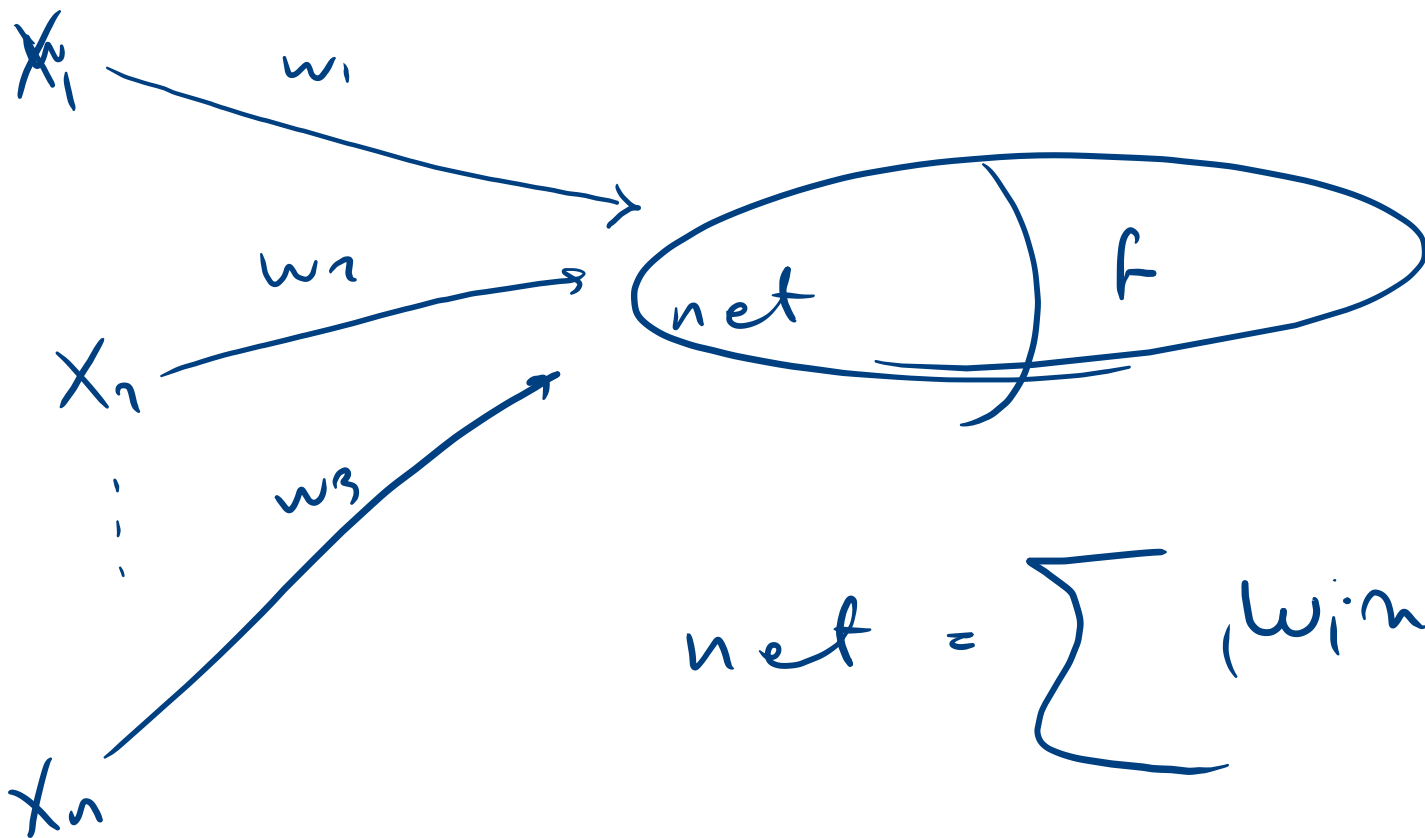
$\left\{ \begin{array}{l} \text{age} \\ \rightarrow X = 3 \\ \text{mat: A (1)} \end{array} \right.$

$$w_1 = 2$$

$$w_2 = 1$$



$$net = \sum w_i x_i + b_i = w_1 x_1 + w_2 x_2 + b = (2 \times 3) + (1 \times 1) = \underline{7}$$

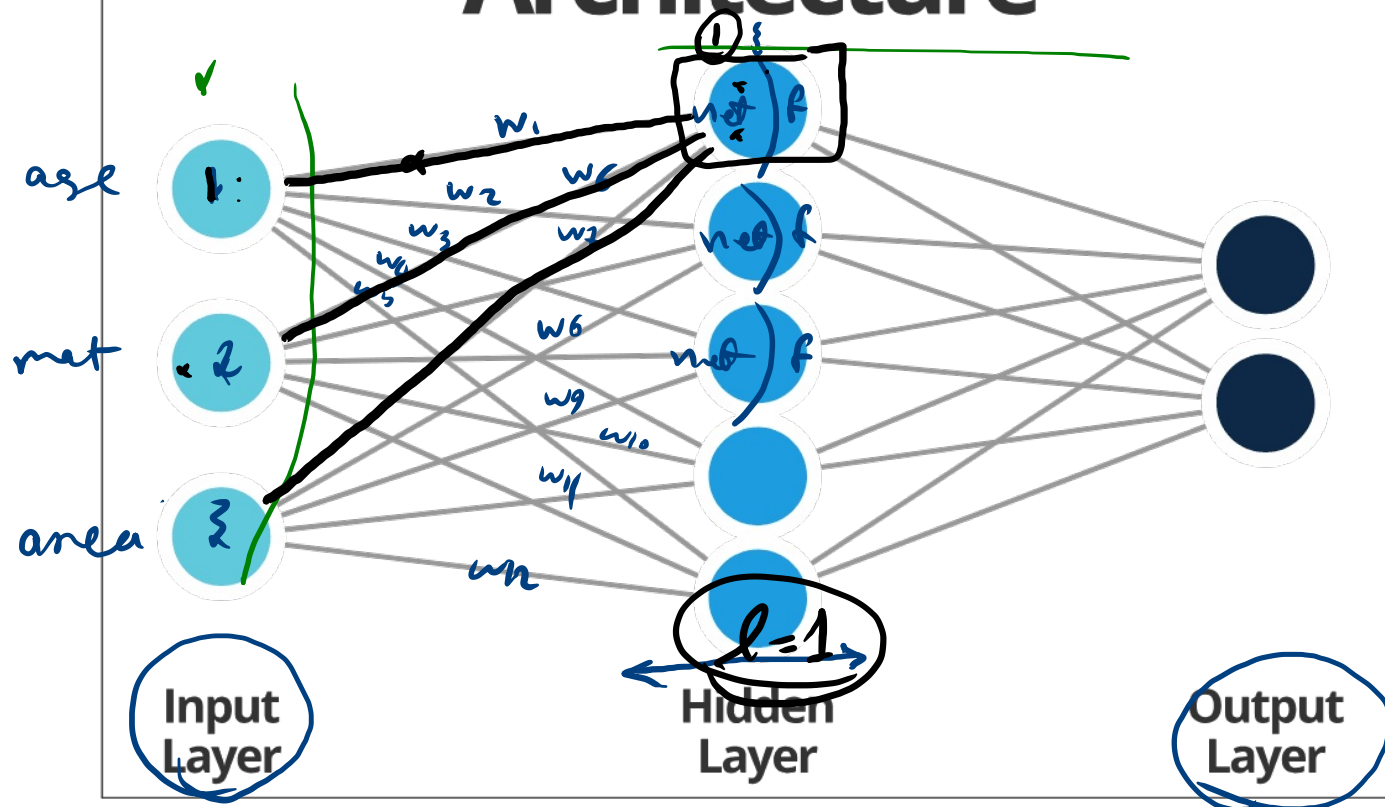


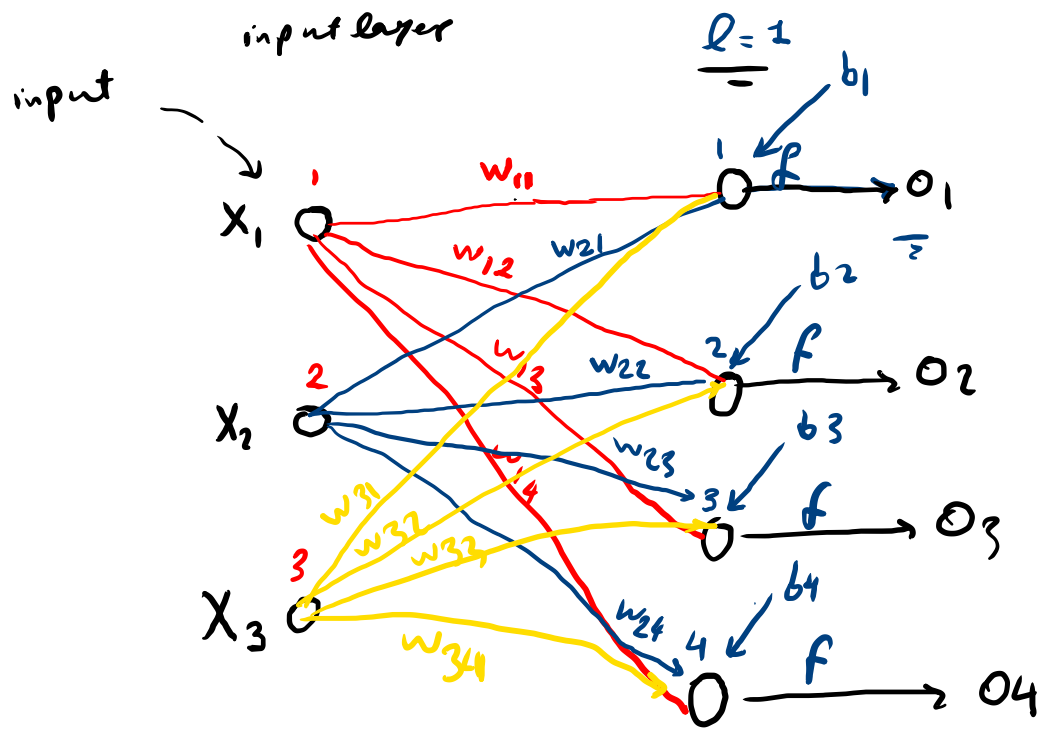
$$\text{net} = \sum (w_i x_i + b_i)$$



| <u>age</u> | <u>mat. area</u> | <u>price</u> |
|------------|------------------|--------------|
|            |                  |              |

# Artificial Neural Network Architecture





$$net_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + b_1$$

$$net_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 + b_2$$

$$net_3 = w_{13}x_1 + w_{23}x_2 + w_{33}x_3 + b_3$$

$$net_4 = w_{14}x_1 + w_{24}x_2 + w_{34}x_3 + b_4$$

$$\begin{bmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \\ w_{13} & w_{23} & w_{33} \\ w_{14} & w_{24} & w_{34} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} net_1 \\ net_2 \\ net_3 \\ net_4 \end{bmatrix} \rightarrow Wx + b = net$$

$$o_1 = f(\text{net}_1)$$

$$o_2 = f(\text{net}_2)$$

$$o_3 = f(\text{net}_3)$$

$$o_4 = f(\text{net}_4)$$

$$\rightarrow O = \begin{bmatrix} o_1 \\ o_2 \\ o_3 \\ o_4 \end{bmatrix} = \begin{bmatrix} f(\text{net}_1) \\ f(\text{net}_2) \\ f(\text{net}_3) \\ f(\text{net}_4) \end{bmatrix} =$$

$$O = f \left( \begin{bmatrix} \text{net}_1 \\ \text{net}_2 \\ \text{net}_3 \\ \text{net}_4 \end{bmatrix} \right) = f(WX + b)$$

$$O = f(wX + b)$$